SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5: Measure network performance using key metrics with simulation tools.
(BL4,BL5)

Assessment Tool Weightage: 30%

Annexure A

Pseudocode Writing Task (Algorithm Design)

1. In a network simulation using Cisco Packet Tracer, a student records total data received as 8000 bits in 4 seconds and observes that 200 packets were sent while 180 packets were received. Write a pseudocode algorithm to calculate the network throughput in bits per second and the packet loss percentage based on the given values. The algorithm should include validation to ensure that time and total packets sent are not zero to avoid errors. It should display both the throughput and packet loss results clearly. Add logic to classify the network as Efficient if throughput is greater than
   1500 bps and packet loss is less than 10%, otherwise classify it as Inefficient. Finally, ensure the algorithm outputs a complete summary of the network performance based on the calculated metrics.
2. An organization has multiple network links connecting its branches. The administrator wants to monitor bandwidth utilization and identify overloaded links. Write a pseudocode algorithm that periodically collects the bandwidth usage of each network link and stores the values in an array or list. Use a loop to continuously monitor all links. Add conditions to detect when bandwidth utilization exceeds 80% and recommend switching traffic to a backup link. Explain how this algorithm supports efficient bandwidth management and prevents network congestion.

3. A company has six branch offices connected through multiple network links. Each link has different bandwidth, delay, and utilization values, which change throughout the day. The network administrator wants to continuously identify the best communication path between the headquarters and each branch based on overall link quality rather than just the shortest distance.Write a pseudocode algorithm that periodically collects the bandwidth, delay, and utilization of every network link and stores the values in suitable arrays or lists. Use a loop to repeat the monitoring process at regular intervals. Calculate a link quality score for each path using the collected metrics and select the path with the highest score. Include conditions to detect when a link becomes overloaded or fails, automatically selecting an alternate path and generating an alert for the administrator. Explain how your algorithm improves network reliability, routing efficiency, and fault tolerance while adapting to changing network conditions.

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Q1. Network Throughput and Packet Loss

Given:

- Total data received = 8000 bits
- Time = 4 seconds
- Packets sent = 200
- Packets received = 180

Formula:

- Throughput = Total Data Received / Time
- Packet Loss % = ((Packets Sent − Packets Received) / Packets Sent) × 100

```
ALGORITHM CalculateNetworkPerformance

BEGIN

   SET dataReceived = 8000
   SET time = 4
   SET packetsSent = 200
   SET packetsReceived = 180

   IF time = 0 THEN
      DISPLAY "Error: Time cannot be zero"
      STOP
   END IF

   IF packetsSent = 0 THEN
      DISPLAY "Error: Total packets sent cannot be zero"
      STOP
   END IF

   throughput = dataReceived / time

   lostPackets = packetsSent - packetsReceived

   packetLoss = (lostPackets / packetsSent) * 100

   DISPLAY "Network Performance"
   DISPLAY "Throughput = ", throughput, " bps"
   DISPLAY "Packet Loss = ", packetLoss, "%"

   IF throughput > 1500 AND packetLoss < 10 THEN
      DISPLAY "Network Classification: Efficient"
   ELSE
      DISPLAY "Network Classification: Inefficient"
   END IF

   DISPLAY "Total Data Received = ", dataReceived, " bits"
   DISPLAY "Total Packets Sent = ", packetsSent
   DISPLAY "Total Packets Received = ", packetsReceived
```

```
    DISPLAY "Packets Lost = ", lostPackets
    DISPLAY "Performance Summary Generated Successfully"

END
```

Output:

```
Network Performance
Throughput = 2000 bps
Packet Loss = 10%

Network Classification: Inefficient

Total Data Received = 8000 bits
Total Packets Sent = 200
Total Packets Received = 180
Packets Lost = 20
Performance Summary Generated Successfully
```

Explanation:
The algorithm first validates the time and number of packets sent to avoid division-by-zero errors. It then calculates throughput and packet loss. Since throughput is **2000 bps**, but packet loss is **10%**, the condition packetLoss < 10 is false. Therefore, the network is classified as **Inefficient**. This directly follows the conditions given in the question.

Q2. Bandwidth Utilization Monitoring

The question requires periodically collecting bandwidth usage, storing it in an array/list, continuously monitoring links, detecting utilization above **80%**, and recommending a backup link.

```
ALGORITHM MonitorBandwidth

BEGIN

    SET numberOfLinks = 5
    CREATE array utilization[5]

    WHILE TRUE DO

        DISPLAY "Checking Network Links..."

        FOR i = 0 TO numberOfLinks - 1 DO

            INPUT utilization[i]

            DISPLAY "Link ", i + 1,
                    " Utilization = ", utilization[i], "%"

            IF utilization[i] > 80 THEN

                DISPLAY "WARNING: Link ", i + 1,
                        " is overloaded"
```

```
            DISPLAY "Recommendation: Switch traffic"
            DISPLAY "to backup link"

        ELSE

            DISPLAY "Link ", i + 1,
                    " is operating normally"

        END IF

    END FOR

    DISPLAY "Bandwidth monitoring completed"
    DISPLAY "Waiting for next monitoring interval..."

    WAIT for next monitoring interval

  END WHILE

END
```

Example Output

```
Checking Network Links...

Link 1 Utilization = 65%
Link 1 is operating normally

Link 2 Utilization = 85%
WARNING: Link 2 is overloaded
Recommendation: Switch traffic to backup link

Link 3 Utilization = 72%
Link 3 is operating normally

Link 4 Utilization = 91%
WARNING: Link 4 is overloaded
Recommendation: Switch traffic to backup link

Link 5 Utilization = 60%
Link 5 is operating normally

Bandwidth monitoring completed
Waiting for next monitoring interval...
```

How it helps

- **Array** stores utilization values of all links.
- **FOR loop** checks every network link.
- **IF condition** detects utilization above 80%.

- An overloaded link triggers a **backup-link recommendation**.
- **WHILE loop** repeats monitoring periodically.
- This helps prevent congestion and supports efficient bandwidth management.

Q3. Best Communication Path Selection

This one is the most important conceptually. The question wants the algorithm to consider **bandwidth, delay and utilization**, calculate an overall **link quality score**, select the highest-scoring path, and automatically switch to an alternate path if a link is overloaded or fails.

```
ALGORITHM SelectBestNetworkPath

  WHILE TRUE DO

    DISPLAY "Collecting Network Link Information..."

    FOR i = 0 TO numberOfPaths - 1 DO

      INPUT bandwidth[i]
      INPUT delay[i]
      INPUT utilization[i]
      INPUT status[i]

      IF status[i] = "FAILED" THEN

        linkScore[i] = 0

        DISPLAY "Path ", i + 1,
            " has FAILED"

      ELSE IF utilization[i] > 80 THEN

        linkScore[i] = 0

        DISPLAY "Path ", i + 1,
            " is OVERLOADED"

      ELSE

        linkScore[i] =
          (bandwidth[i] / 100)
          + (100 - delay[i]) / 100
          + (100 - utilization[i]) / 100

        DISPLAY "Path ", i + 1,
            " Score = ", linkScore[i]

      END IF
```

```
        END FOR

        SET highestScore = -1
        SET bestPath = -1

        FOR i = 0 TO numberOfPaths - 1 DO

            IF linkScore[i] > highestScore THEN

                highestScore = linkScore[i]
                bestPath = i + 1

            END IF

        END FOR

        IF bestPath = -1 THEN

            DISPLAY "ALERT: No suitable network path available"

        ELSE

            DISPLAY "Best Path = Path ", bestPath
            DISPLAY "Highest Link Quality Score = ",
                    highestScore

        END IF

        DISPLAY "Network monitoring completed"
        DISPLAY "Repeating after regular interval..."

        WAIT for next monitoring interval

    END WHILE

END
```

Example Output

```
Collecting Network Link Information...

Path 1 Score = 2.15
Path 2 Score = 2.48
Path 3 is OVERLOADED
Path 4 Score = 2.31
Path 5 has FAILED
Path 6 Score = 2.65

Best Path = Path 6
```

Highest Link Quality Score = 2.65

Network monitoring completed
Repeating after regular interval...